

Biswadeep Tewari

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Durgapur, West Bengal, India | Open to Relocation

Education

B.Tech, Computer Science Engineering - MAKAUT, West Bengal *Expected: 07/2027*
Relevant coursework: Machine Learning, Data Structures & Algorithms, Distributed Systems, Computer Vision

Technical Skills

Languages	Python, SQL, Kotlin, JavaScript/TypeScript, Dart (Flutter), C++
Systems	Concurrency & synchronization (mutex/lock design, async), distributed multi-process architecture (IPC, subprocess isolation), CUDA/VRAM scheduling, WebAssembly
ML / AI	PyTorch, TensorFlow, llama.cpp, Stable Diffusion, LangChain, MediaPipe, OpenCV, SAM2, GroundingDINO, SegFormer, Whisper, Hugging Face
Cloud / Infra	Docker, Kubernetes, Redis, Celery, FastAPI, async, OpenTelemetry, GitHub Actions
Google Cloud	BigQuery, Vertex AI, GKE, Pub/Sub, Cloud SQL, Cloud Run, Gemini API, Agent Development Kit

Projects

Neural Citadel - AI Model Serving Platform *2025 – Present*
Python, FastAPI, PyQt6, Flutter, llama.cpp, Stable Diffusion, SAM2, GroundingDINO, SegFormer, Whisper, async, GitHub Actions | github.com/RajTewari01/neural_citadel

Runs 40+ AI models in production on a single consumer GPU instead of needing enterprise hardware.

- Built a subprocess isolation architecture around a centralised GPU mutex (ResourceManager) for cross-process synchronization, almost completely reclaiming VRAM when a process exits and fixing the CUDA memory fragmentation that breaks most long-running multi-model setups
- Orchestrated 40+ heterogeneous models (diffusion, LLMs, segmentation, speech) across 12 isolated engines on a single GTX 1650 (4 GB VRAM), cutting hardware cost by about 94% against comparable cloud GPU instances
- Designed two IPC modes over stdin/stdout (batched for diffusion pipelines, a persistent loop for LLMs) behind a 13-router FastAPI gateway that handles concurrent requests with async, running in production since 2025 with no manual intervention
- Built a 6-stage computer vision pipeline (GroundingDINO -> SAM2 -> SegFormer -> CatVTON -> SCHP -> background inpainting) and wrote it up as a research preprint on Zenodo, indexed on OpenAIRE with 341 views and 306 downloads

Social Management Agent - Agentic Automation Platform *2025 – Present*
LangChain, FastAPI, Redis, Celery, OpenTelemetry, Pydantic, Flutter, Kotlin

Automates Instagram, X, LinkedIn, and Facebook for creators and small businesses priced out of Western SaaS tools.

- Building a bring-your-own-key multi-agent orchestration platform so users supply their own API keys instead of paying for a subscription
- Built a distributed task queue on Celery and Redis to schedule and run agents concurrently and reliably, instrumented end to end with OpenTelemetry and a live Grafana dashboard tracking GPU, VRAM, API latency, and engagement metrics
- Trained a VotingClassifier ensemble (TF-IDF, logistic regression, linear SVM, multinomial naive Bayes) on a 300k-row synthetic dataset for spam detection, paired with a custom regex rule builder for content filtering

Spatial Tracer - Gesture Control Engine *2025 – Present*
Python, Flutter, Kotlin, MediaPipe, Three.js, WebAssembly, GitHub Actions, MIT Licensed | github.com/RajTewari01/spatial-tracer

Touchless computer control from any camera, no specialized hardware or training data needed.

- Built a deterministic kinematic engine that turns live camera input into real OS events using geometric heuristics on 21 3D hand joints and 478 facial landmarks, with no ML classifier or training data, running at native camera frame rate
- Shipped it across three independent platforms: a Python/PyQt5 desktop app, an Android/Kotlin app using AccessibilityService for system-wide input injection, and a WebAssembly/Three.js web app with a 4000-particle physics

simulation, all with CI/CD through GitHub Actions

Certifications

Google Cloud (30+ Skill Badges, 04/2026): hands-on labs across ML APIs, AI Agents with ADK, Gemini Enterprise, BigQuery, Kubernetes, and cloud networking.

Microsoft Azure (06/2026): speech and text-analysis generative AI agents with Azure AI Foundry, plus three GitHub Actions CI/CD certifications.

NVIDIA (04/2026): Deploy Faster Generative AI Models. **Deloitte Australia** (06/2026): Data Analytics Job Simulation (Forage).

Leadership & Community

Google Campus Ambassador - MAKAUT *2026*

GSSoC Ambassador, Contributor & Chapter Lead - GirlScript Summer of Code *2026*

Participant - Microsoft Agents League Hackathon (AI Agents track) *06/2026*